Kairui Zhang

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PROFILE

My research interests lie in multi-modal learning, encompassing computer vision and natural language processing. I am passionate about utilizing neural networks and reinforcement learning methods to solve problems.

EDUCATION

University of Bristol 2025 QS Rank 54 09/2024 - Present
PhD in Engineering Mathematics
University of Bristol 09/2022 - 09/2023
MSc in Engineering Mathematics
Hohai University University of 211 Project; 985 Project Innovation Platform 09/2018 - 06/2022
BSc in Robotics Engineering

PUBLICATIONS

[1] K. Zhang and M. Lewis, "Evaluating CLIP's understanding on relationships in a blocks world," in *2023 IEEE international conference on big data (Big data)*, IEEE, 2023.

[2] P. Zhang, C. Wei, K. Zhang, and Y. Ouyang, "Self-learning approach to control parameter adjustment for quadcopter landing on a moving platform," *CAAI Transactions on Intelligent Systems*, vol. 17, no. 5, pp. 931–940, 2022. DOI: 10.11992/tis.202107040.

AWARDS

Kaggle Competition *Stable Diffusion - Image to Prompts* silver medal 05/2023
BRISTOL BIO-HACKATHON First prize 11/2022
Hohai University School Scholarship 05/2021

PROJECT EXPERIENCE

Compositional Generalization in Text-to-Image Synthesis. 11/2022 - 11/2023

- Investigated the performance of text-image generation models in the relationship binding problem.
- Replicated the stable diffusion with high-performance computer and Hugging Face packages.
- Conducted a comprehensive evaluation encompassing CLIP and its various fine-tuned iterations on a relational dataset.
- Applied an ensemble method to CLIP, incorporating distinct model features, including Symbolic methods, BERT and the T5 model.
- Employed two learning strategies, an auxiliary classifier and a compositional loss, to train the CLIP model.

Scoring Protein-Protein Interactions 01/2023 - 03/2023

- Predicted probability that two proteins can interact by machine learning method.
- Investigated the correlation between energy-based metrics and structure-based metrics.
- Working in a team of 4, I practiced communication skills and cooperation.
- Worked with external stakeholders to deliver on their vision.

Autonomous Landing of UAV for UAV-UGV Cooperation 04/2021 - 06/2022

- Supported by the National Natural Science Foundation of China; the Fundamental Research Funds for the Central Universities.
- Collected a dataset in ROS by OpenCV and traditional computer vision methods.
- Detected the landmark based on YOLOX and altitude estimation method.
- Improved detection accuracy by 4.17% through boosting the dataset with DCGAN.
- Landed the drone on a moving UGV by DDPG.
- Deployed these algorithms in real-world experiment.

Gesture Recognition Based UGV Navigation 03/2021 - 04/2021

- Trained a Shuffle Net to navigate UGV by Gesture Recognition.
- Implemented PID controller and advanced controller in STM32.
- Deployed the model in RaspberryPi.

WORK EXPERIENCE

Volex Interconnect Systems (Suzhou) Co., Ltd 05/2024 - 09/2024

Firmware Engineer

- Designed asynchronous cable test software architecture in C#.
- Refactored specific test services.
- Managed Git server.

Engineering Research Center of Dredging Technology, Ministry of Education 09/2020 - 06/2022

Undergraduate Research Assistant | Program Directors: Changyun Wei

- Performed gesture recognition with Shuffle Net in a designed UGV.
- Implemented object detection and control method in a UAV.
- Demonstrated that I can work well as part of a team and communicate technical work clearly.

Changzhou Guli High-End Equipment Innovation Center 06/2021 - 07/2021

Robotics Software Engineer Intern

- Implemented path planning for SCOUT Mini robot by ROS.
- Utilized RPLIDAR to obstacle avoidance.
- Increased accuracy by Intel RealSense depth camera D435i with OpenCV in C++.

SKILLS

IT Skills

- Programming: proficient in Python, experienced in C/C++, MATLAB, Bash and C#.
- Deep Learning: proficient in PyTorch, experienced in TensorFlow1.14.0, Kares, PaddlePaddle
- Data Processing: Matplotlib, NumPy, Pandas, scikit-learn, SciPy, xgboost, polars
- Applications: Slurm, Git, VS Code, Vim, Jupyter, LaTeX, Microsoft Office
- Operating Systems: familiar with Linux (Ubuntu, CentOS), Windows
- Chinese National Computer Rank Examination Grade II (C++) and Grade III (computer network)
- Frontend: HTML5, CSS, JavaScript and Vue
- Database: MySQL

Languages

Mandarin Chinese (native), English (IELTS 6.5, fluent working English), Japanese (basic)